

Draft

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1 Impact of frequency shift on amplitude

We have derived that the numerical angular frequency is given by

$$\omega_{\text{ADI}}(k_x) = \frac{2}{\Delta t} \arctan\left(S \sin \frac{k_x \Delta x}{2}\right), \quad (1)$$

thus the frequency shift is given by

$$\frac{\omega_{\text{ADI}}(k_x)}{\omega_0} = \frac{2 \arctan[S \sin(k \Delta z / 2)]}{S k \Delta z}. \quad (2)$$

Consider an input source whose temporal profile is given by

$$s(t) = \alpha \exp\left[-\frac{(t - t_0)^2}{2T_P^2}\right] \sin(\omega_d t), \quad (3)$$

compute the Fourier-transform by $\hat{s}(\omega) = \int_{-\infty}^{\infty} s(t) e^{-i\omega t} dt$,

$$\hat{s}(\omega) = \frac{\alpha \sqrt{2\pi} T_P}{2i} e^{-i\omega t_0} \left[e^{i\omega_d t_0} e^{-\frac{1}{2} T_P^2 (\omega - \omega_d)^2} - e^{-i\omega_d t_0} e^{-\frac{1}{2} T_P^2 (\omega + \omega_d)^2} \right]. \quad (4)$$

Consider ω_{ADI} that is close to ω_d , so we have

$$e^{-\frac{1}{2} T_P^2 (\omega_{\text{ADI}} + \omega_d)^2} \ll e^{-\frac{1}{2} T_P^2 (\omega_{\text{ADI}} - \omega_d)^2}, \quad (5)$$

thus

$$\hat{s}(\omega_{\text{ADI}}) \approx \frac{\alpha \sqrt{2\pi} T_P}{2i} e^{-i(\omega_{\text{ADI}} - \omega_d) t_0} e^{-\frac{1}{2} T_P^2 (\omega_{\text{ADI}} - \omega_d)^2}. \quad (6)$$

Taking the magnitude,

$$|\hat{s}(\omega_{\text{ADI}})| \approx \frac{\alpha \sqrt{2\pi} T_P}{2} e^{-\frac{1}{2} T_P^2 (\omega_{\text{ADI}} - \omega_d)^2} \propto e^{-\frac{1}{2} T_P^2 (\omega_{\text{ADI}} - \omega_d)^2}. \quad (7)$$

2 Source term analysis

2.1 Electric field source terms

Use the conventions $\delta_x^{E,2} = \delta_x^H \delta_x^E$ and $\delta_x^{H,2} = \delta_x^E \delta_x^H$. Let $S_{E,1}^n$ be the electric field source term in between n and $n + \frac{1}{2}$, and $S_{E,2}^n$ be that in between $n + \frac{1}{2}$ and $n + 1$.

$$(1 - \gamma \delta_x^{E,2}) E_z^{n+\frac{1}{2}} = E_z^n + C_b \delta_x^H H_y^n + S_{E,1}^n \quad (8)$$

$$H_y^{n+\frac{1}{2}} = H_y^n + D_b \delta_x^E E_z^{n+\frac{1}{2}} \quad (9)$$

$$E_z^{n+1} = E_z^{n+\frac{1}{2}} + C_b \delta_x^H H_y^{n+\frac{1}{2}} + S_{E,2}^n \quad (10)$$

$$H_y^{n+1} = H_y^{n+\frac{1}{2}} + D_b \delta_x^E E_z^{n+\frac{1}{2}} \quad (11)$$

Shifting back equations (10) and (11) one full step gives

$$E_z^n = E_z^{n-\frac{1}{2}} + C_b \delta_x^H H_y^{n-\frac{1}{2}} + S_{E,2}^{n-1} \quad (12)$$

$$H_y^n = H_y^{n-\frac{1}{2}} + D_b \delta_x^E E_z^{n-\frac{1}{2}} \quad (13)$$

To eliminate H_y from equations (8)–(10), substitute equation (9) into equation (10):

$$E_z^{n+1} = E_z^{n+\frac{1}{2}} + C_b \delta_x^H H_y^n + \gamma \delta_x^{E,2} E_z^{n+\frac{1}{2}} + S_{E,2}^n. \quad (14)$$

Equation (8) gives

$$C_b \delta_x^H H_y^n = (1 - \gamma \delta_x^{E,2}) E_z^{n+\frac{1}{2}} - E_z^n - S_{E,1}^n. \quad (15)$$

Using $\gamma \delta_x^{E,2} = C_b D_b \delta_x^H \delta_x^E$, equations (14) and (15) reduce to

$$E_z^{n+\frac{1}{2}} = \frac{1}{2} (E_z^{n+1} + E_z^n + S_{E,1}^n - S_{E,2}^n). \quad (16)$$

Shifting back one full step gives

$$E_z^{n-\frac{1}{2}} = \frac{1}{2} (E_z^n + E_z^{n-1} + S_{E,1}^{n-1} - S_{E,2}^{n-1}). \quad (17)$$

Next, eliminate H_y using equations (8), (12), and (13). Applying $C_b \delta_x^H$ to both sides of equation (13) gives

$$C_b \delta_x^H H_y^n = C_b \delta_x^H H_y^{n-\frac{1}{2}} + \gamma \delta_x^{E,2} E_z^{n-\frac{1}{2}}. \quad (18)$$

Eliminate $C_b \delta_x^H H_y^{n-\frac{1}{2}}$ by equation (12) gives

$$C_b \delta_x^H H_y^n = E_z^n - (1 - \gamma \delta_x^{E,2}) E_z^{n-\frac{1}{2}} - S_{E,2}^{n-1}. \quad (19)$$

Substituting into equation (8) to eliminate H_y :

$$(1 - \gamma \delta_x^{E,2}) (E_z^{n+\frac{1}{2}} + E_z^{n-\frac{1}{2}}) = 2E_z^n + S_{E,1}^n - S_{E,2}^{n-1}. \quad (20)$$

Finally, substitute equations (16) and (17) into equation (20) to eliminate the half-step values of E_z :

$$(1 - \gamma \delta_x^{E,2}) E_z^{n+1} = 2(1 + \gamma \delta_x^{E,2}) E_z^n - (1 - \gamma \delta_x^{E,2}) E_z^{n-1} + (1 + \gamma \delta_x^{E,2}) S_{E,1}^n - (1 - \gamma \delta_x^{E,2}) S_{E,1}^{n-1} + (1 - \gamma \delta_x^{E,2}) S_{E,2}^n - (1 + \gamma \delta_x^{E,2}) S_{E,2}^{n-1}. \quad (21)$$

Recall that $\gamma = C_b D_b = c^2 \Delta t^2 / 4$, then equation (21) results in,

$$\frac{E_z^{n+1} - 2E_z^n + E_z^{n-1}}{\Delta t^2} = c^2 \delta_x^{E,2} \left(\frac{E_z^{n+1} + 2E_z^n + E_z^{n-1}}{4} \right) + \frac{1}{\Delta t^2} \left[(1 + \gamma \delta_x^{E,2}) S_{E,1}^n - (1 - \gamma \delta_x^{E,2}) S_{E,1}^{n-1} + (1 - \gamma \delta_x^{E,2}) S_{E,2}^n - (1 + \gamma \delta_x^{E,2}) S_{E,2}^{n-1} \right]. \quad (22)$$

Suppose the same source is applied in both substeps, with $S_{E,1}^n = S_{E,2}^n = \frac{\Delta t}{2} S_E^{n+\frac{1}{2}}$ and $S_{E,1}^{n-1} = S_{E,2}^{n-1} = \frac{\Delta t}{2} S_E^{n-\frac{1}{2}}$. Then equation (22) becomes

$$\frac{E_z^{n+1} - 2E_z^n + E_z^{n-1}}{\Delta t^2} = c^2 \delta_x^{E,2} \left(\frac{E_z^{n+1} + 2E_z^n + E_z^{n-1}}{4} \right) + \frac{S_E^{n+\frac{1}{2}} - S_E^{n-\frac{1}{2}}}{\Delta t}. \quad (23)$$

Assume $S_E^{n+a} = S_E(t_n + a\Delta t)$ and $F_E^n = (S_E)_t(t_n)$. The source approximation in equation (23) has the Taylor expansion

$$\frac{S_E^{n+\frac{1}{2}} - S_E^{n-\frac{1}{2}}}{\Delta t} = F_E^n + \frac{\Delta t^2}{24} (S_E)_{ttt}^n + O(\Delta t^4), \quad \text{error} = \frac{\Delta t^2}{24} (S_E)_{ttt}^n + O(\Delta t^4). \quad (24)$$

Thus the two-substep midpoint-source case is second-order accurate for $F_E^n = (S_E)_t(t_n)$.

Alternatively, suppose the source is added only once in the first substep and evaluated at the midpoint, with $S_{E,1}^n = \Delta t S_E^{n+\frac{1}{2}}$, $S_{E,2}^n = 0$, $S_{E,1}^{n-1} = \Delta t S_E^{n-\frac{1}{2}}$, and $S_{E,2}^{n-1} = 0$. Then equation (22) becomes

$$\frac{E_z^{n+1} - 2E_z^n + E_z^{n-1}}{\Delta t^2} = c^2 \delta_x^{E,2} \left(\frac{E_z^{n+1} + 2E_z^n + E_z^{n-1}}{4} \right) + \frac{1}{\Delta t} \left[(1 + \gamma \delta_x^{E,2}) S_E^{n+\frac{1}{2}} - (1 - \gamma \delta_x^{E,2}) S_E^{n-\frac{1}{2}} \right]. \quad (25)$$

The source approximation in equation (25) has the Taylor expansion

$$F_E^n + \frac{2\gamma}{\Delta t} \delta_x^{E,2} S_E^n + \frac{\Delta t^2}{24} (S_E)_{ttt}^n + O(\Delta t^3). \quad (26)$$

Thus the single midpoint-source case is first-order accurate for a spatially varying S_E , and second-order when $\delta_x^{E,2} S_E = 0$.

Alternatively, suppose the first source is evaluated at the first quarter-step and the second source at the third quarter-step: $S_{E,1}^n = \frac{\Delta t}{2} S_E^{n+\frac{1}{4}}$, $S_{E,2}^n = \frac{\Delta t}{2} S_E^{n+\frac{3}{4}}$, $S_{E,1}^{n-1} = \frac{\Delta t}{2} S_E^{n-\frac{3}{4}}$, and $S_{E,2}^{n-1} = \frac{\Delta t}{2} S_E^{n-\frac{1}{4}}$. Then equation (22) becomes

$$\frac{E_z^{n+1} - 2E_z^n + E_z^{n-1}}{\Delta t^2} = c^2 \delta_x^{E,2} \left(\frac{E_z^{n+1} + 2E_z^n + E_z^{n-1}}{4} \right) + \frac{1}{2\Delta t} \left[(1 + \gamma \delta_x^{E,2}) S_E^{n+\frac{1}{4}} - (1 - \gamma \delta_x^{E,2}) S_E^{n-\frac{3}{4}} + (1 - \gamma \delta_x^{E,2}) S_E^{n+\frac{3}{4}} - (1 + \gamma \delta_x^{E,2}) S_E^{n-\frac{1}{4}} \right]. \quad (27)$$

The source approximation in equation (27) has the Taylor expansion

$$F_E^n - \frac{\gamma}{2} \delta_x^{E,2} F_E^n + \frac{7\Delta t^2}{96} (S_E)_{ttt}^n + O(\Delta t^4), \quad \text{error} = -\frac{\gamma}{2} \delta_x^{E,2} F_E^n + \frac{7\Delta t^2}{96} (S_E)_{ttt}^n + O(\Delta t^4). \quad (28)$$

Since $\gamma = O(\Delta t^2)$, the quarter-step source case is also second-order accurate.

Assume $S_E(x, t) = \hat{S}_E e^{i(\omega t - kx)}$, and define $\theta = \omega \Delta t$, $\delta_x^{E,2} e^{-ikx} = \lambda_x e^{-ikx}$, and $q = -\gamma \lambda_x \geq 0$. Substituting into the source term on the right-hand side of equation (23) gives

$$\begin{aligned} F_{E,\text{mid},2}^n &= \frac{S_E^{n+\frac{1}{2}} - S_E^{n-\frac{1}{2}}}{\Delta t} \\ &= \frac{\hat{S}_E e^{i(\omega t_n - kx)}}{\Delta t} (e^{i\theta/2} - e^{-i\theta/2}) \\ &= \frac{2i\hat{S}_E}{\Delta t} \sin\left(\frac{\theta}{2}\right) e^{i(\omega t_n - kx)}, \end{aligned} \quad (29)$$

so $\hat{F}_{E,\text{mid},2} = \frac{2i\hat{S}_E}{\Delta t} \sin\left(\frac{\theta}{2}\right)$. Substituting into the source term on the right-hand side of equation (25) gives

$$\begin{aligned} F_{E,\text{mid},1}^n &= \frac{1}{\Delta t} \left[(1 + \gamma \delta_x^{E,2}) S_E^{n+\frac{1}{2}} - (1 - \gamma \delta_x^{E,2}) S_E^{n-\frac{1}{2}} \right] \\ &= \frac{\hat{S}_E e^{i(\omega t_n - kx)}}{\Delta t} \left[(1 - q) e^{i\theta/2} - (1 + q) e^{-i\theta/2} \right] \\ &= \frac{2\hat{S}_E}{\Delta t} \left[i \sin\left(\frac{\theta}{2}\right) - q \cos\left(\frac{\theta}{2}\right) \right] e^{i(\omega t_n - kx)}. \end{aligned} \quad (30)$$

so $\hat{F}_{E,\text{mid},1} = \frac{2\hat{S}_E}{\Delta t} [i \sin(\frac{\theta}{2}) - q \cos(\frac{\theta}{2})]$. Substituting into the source term on the right-hand side of equation (27) gives

$$\begin{aligned} F_{E,\text{quarter}}^n &= \frac{1}{2\Delta t} \left[(1 + \gamma \delta_x^{E,2}) S_E^{n+\frac{1}{4}} - (1 - \gamma \delta_x^{E,2}) S_E^{n-\frac{3}{4}} + (1 - \gamma \delta_x^{E,2}) S_E^{n+\frac{3}{4}} - (1 + \gamma \delta_x^{E,2}) S_E^{n-\frac{1}{4}} \right] \\ &= \frac{\hat{S}_E e^{i(\omega t_n - kx)}}{2\Delta t} \left[(1 - q) e^{i\theta/4} - (1 + q) e^{-3i\theta/4} + (1 + q) e^{3i\theta/4} - (1 - q) e^{-i\theta/4} \right] \\ &= \frac{i\hat{S}_E}{\Delta t} \left[(1 - q) \sin\left(\frac{\theta}{4}\right) + (1 + q) \sin\left(\frac{3\theta}{4}\right) \right] e^{i(\omega t_n - kx)}. \end{aligned} \quad (31)$$

so $\hat{F}_{E,\text{quarter}} = \frac{i\hat{S}_E}{\Delta t} [(1 - q) \sin(\frac{\theta}{4}) + (1 + q) \sin(\frac{3\theta}{4})]$. Therefore the magnitude ratios among the midpoint and quarter-step source terms are

$$\frac{|\hat{F}_{E,\text{mid},1}|}{|\hat{F}_{E,\text{mid},2}|} = \sqrt{1 + q^2 \cot^2\left(\frac{\theta}{2}\right)}, \quad (32)$$

$$\frac{|\hat{F}_{E,\text{quarter}}|}{|\hat{F}_{E,\text{mid},2}|} = \left| \cos\left(\frac{\theta}{4}\right) + \frac{q \cos(\frac{\theta}{2})}{2 \cos(\frac{\theta}{4})} \right|, \quad (33)$$

$$\frac{|\hat{F}_{E,\text{quarter}}|}{|\hat{F}_{E,\text{mid},1}|} = \frac{|(1 - q) \sin(\frac{\theta}{4}) + (1 + q) \sin(\frac{3\theta}{4})|}{2\sqrt{\sin^2(\frac{\theta}{2}) + q^2 \cos^2(\frac{\theta}{2})}}. \quad (34)$$

For a spatially uniform source, $q = 0$, both midpoint placements give the same magnitude, and the quarter-step magnitude is smaller by the factor

$$\frac{|\hat{F}_{E,\text{mid},1}|}{|\hat{F}_{E,\text{mid},2}|} = 1, \quad \frac{|\hat{F}_{E,\text{quarter}}|}{|\hat{F}_{E,\text{mid},2}|} = \frac{|\hat{F}_{E,\text{quarter}}|}{|\hat{F}_{E,\text{mid},1}|} = \left| \cos\left(\frac{\omega \Delta t}{4}\right) \right|. \quad (35)$$

2.2 Method of manufactured solution validation for electric source

To test the electric-source placements of the previous section with a genuinely nonzero forcing, consider the one-dimensional wave equation

$$E_{tt} - c^2 E_{xx} = f(x, t), \quad x \in [0, L], \quad (36)$$

with PEC boundary condition, $E(0, t) = E(L, t) = 0$. Choose a smooth manufactured solution,

$$E_{\text{MMS}}(x, t) = \sin(kx) \cos(\omega t), \quad k = \frac{2\pi}{L}, \quad \omega = \frac{3\pi c}{L}, \quad (37)$$

where we set $\omega \neq ck$ in order to have a non-zero f .

Indeed,

$$E_{tt} = -\omega^2 \sin(kx) \cos(\omega t), \quad E_{xx} = -k^2 \sin(kx) \cos(\omega t), \quad (38)$$

so

$$f(x, t) = (c^2 k^2 - \omega^2) \sin(kx) \cos(\omega t) = -\frac{5\pi^2 c^2}{L^2} \sin(kx) \cos(\omega t). \quad (39)$$

In the ADI update, $\partial_t E = \varepsilon^{-1} \partial_x H + S_E$ with $F_E = (S_E)_t$ the wave-equation source of (22). Matching $F_E = f$ therefore gives the continuous electric soft source

$$S_E(x, t) = \frac{c^2 k^2 - \omega^2}{\omega} \sin(kx) \sin(\omega t). \quad (40)$$

From (37), we have:

$$E(x, 0) = \sin(kx), \quad E_t(x, 0) = 0, \quad (41)$$

and the compatible magnetic field is $H(x, 0) = 0$ (since $S_E(x, 0) = 0$ as well). The wavenumber $k = 2\pi/L$ satisfies the PEC conditions.

To analyse how the magnitude will scale with respect to the CFL numbers, we model the system as a forced oscillation system with natural frequency $\omega_{\text{ADI}}(k)$ and a forcing function whose frequency is $\omega_d = \omega$ from (37).

We model the system as a simple undamped forced oscillator,

$$a'' + \omega_{\text{ADI}}^2 a = F_E(t). \quad (42)$$

First, we consider $F_E(t) = F_{E, \text{mid}, 2}$ from (29). With the manufactured soft source (40),

$$\begin{aligned} F_{E, \text{mid}, 2} &= \frac{S_E^{n+\frac{1}{2}} - S_E^{n-\frac{1}{2}}}{\Delta t} \\ &= \frac{c^2 k^2 - \omega^2}{\omega} \sin(kx) \frac{\sin(\omega(t_n + \frac{1}{2}\Delta t)) - \sin(\omega(t_n - \frac{1}{2}\Delta t))}{\Delta t} \\ &= \frac{c^2 k^2 - \omega^2}{\omega} \sin(kx) \frac{2}{\Delta t} \sin\left(\frac{\omega\Delta t}{2}\right) \cos(\omega t_n). \end{aligned} \quad (43)$$

Thus corresponding system is

$$\begin{aligned} a'' + \omega_{\text{ADI}}^2 a &= \frac{2}{\Delta t} \sin\left(\frac{\omega\Delta t}{2}\right) \cdot \frac{c^2 k^2 - \omega^2}{\omega} \cos(\omega t), \\ a(0) &= 1, \\ a'(0) &= 0. \end{aligned} \quad (44)$$

Solving this system yields

$$\begin{aligned} a(t) &= (1 - A_p) \cos(\omega_{\text{ADI}} t) + A_p \cos(\omega t), \\ A_p &= \text{sinc}\left(\frac{\omega\Delta t}{2}\right) \frac{c^2 k^2 - \omega^2}{\omega_{\text{ADI}}^2 - \omega^2}. \end{aligned} \quad (45)$$

Next, consider $F_E(t) = F_{E, \text{mid}, 1}$ from (30). On the mode $\sin(kx)$ one has $\delta_x^{E, 2} \sin(kx) = \lambda_x \sin(kx)$ and $q = -\gamma\lambda_x$, so with (40),

$$\begin{aligned} F_{E, \text{mid}, 1} &= \frac{1}{\Delta t} \left[(1 + \gamma\delta_x^{E, 2}) S_E^{n+\frac{1}{2}} - (1 - \gamma\delta_x^{E, 2}) S_E^{n-\frac{1}{2}} \right] \\ &= \frac{c^2 k^2 - \omega^2}{\omega\Delta t} \sin(kx) \left[(1 - q) \sin(\omega(t_n + \frac{1}{2}\Delta t)) - (1 + q) \sin(\omega(t_n - \frac{1}{2}\Delta t)) \right] \\ &= \frac{2(c^2 k^2 - \omega^2)}{\omega\Delta t} \sin(kx) \left[\sin\left(\frac{\omega\Delta t}{2}\right) \cos(\omega t_n) - q \cos\left(\frac{\omega\Delta t}{2}\right) \sin(\omega t_n) \right]. \end{aligned} \quad (46)$$

The corresponding modal system is

$$\begin{aligned} a'' + \omega_{\text{ADI}}^2 a &= \frac{2(c^2 k^2 - \omega^2)}{\omega \Delta t} \left[\sin\left(\frac{\omega \Delta t}{2}\right) \cos(\omega t) - q \cos\left(\frac{\omega \Delta t}{2}\right) \sin(\omega t) \right], \\ a(0) &= 1, \\ a'(0) &= 0. \end{aligned} \quad (47)$$

Writing the drive as $A \cos(\omega t) + B \sin(\omega t)$ with

$$A = \frac{2(c^2 k^2 - \omega^2)}{\omega \Delta t} \sin\left(\frac{\omega \Delta t}{2}\right), \quad B = -\frac{2(c^2 k^2 - \omega^2)}{\omega \Delta t} q \cos\left(\frac{\omega \Delta t}{2}\right), \quad (48)$$

the solution is

$$\begin{aligned} a(t) &= \left(1 - \frac{A}{\omega_{\text{ADI}}^2 - \omega^2}\right) \cos(\omega_{\text{ADI}} t) - \frac{\omega}{\omega_{\text{ADI}}} \frac{B}{\omega_{\text{ADI}}^2 - \omega^2} \sin(\omega_{\text{ADI}} t) \\ &\quad + \frac{A}{\omega_{\text{ADI}}^2 - \omega^2} \cos(\omega t) + \frac{B}{\omega_{\text{ADI}}^2 - \omega^2} \sin(\omega t). \end{aligned} \quad (49)$$

Finally, consider $F_E(t) = F_{E,\text{quarter}}$ from (31). With (40),

$$\begin{aligned} F_{E,\text{quarter}} &= \frac{1}{2\Delta t} \left[(1 + \gamma \delta_x^{E,2}) S_E^{n+\frac{1}{4}} - (1 - \gamma \delta_x^{E,2}) S_E^{n-\frac{3}{4}} \right. \\ &\quad \left. + (1 - \gamma \delta_x^{E,2}) S_E^{n+\frac{3}{4}} - (1 + \gamma \delta_x^{E,2}) S_E^{n-\frac{1}{4}} \right] \\ &= \frac{c^2 k^2 - \omega^2}{\omega \Delta t} \sin(kx) \left[(1 - q) \sin\left(\frac{\omega \Delta t}{4}\right) + (1 + q) \sin\left(\frac{3\omega \Delta t}{4}\right) \right] \cos(\omega t_n). \end{aligned} \quad (50)$$

The corresponding modal system is

$$\begin{aligned} a'' + \omega_{\text{ADI}}^2 a &= \frac{c^2 k^2 - \omega^2}{\omega \Delta t} \left[(1 - q) \sin\left(\frac{\omega \Delta t}{4}\right) + (1 + q) \sin\left(\frac{3\omega \Delta t}{4}\right) \right] \cos(\omega t), \\ a(0) &= 1, \\ a'(0) &= 0. \end{aligned} \quad (51)$$

Solving yields

$$\begin{aligned} a(t) &= (1 - A_p) \cos(\omega_{\text{ADI}} t) + A_p \cos(\omega t), \\ A_p &= \frac{c^2 k^2 - \omega^2}{(\omega_{\text{ADI}}^2 - \omega^2) \omega \Delta t} \left[(1 - q) \sin\left(\frac{\omega \Delta t}{4}\right) + (1 + q) \sin\left(\frac{3\omega \Delta t}{4}\right) \right]. \end{aligned} \quad (52)$$

Relative to mid,2, the particular amplitudes satisfy the same magnitude ratios as (32)–(33).

2.3 Magnetic field source terms

Let $S_{H,1}^n$ be the magnetic field source term in between n and $n + \frac{1}{2}$, and $S_{H,2}^n$ be that in between $n + \frac{1}{2}$ and $n + 1$.

$$(1 - \gamma \delta_x^{E,2}) E_z^{n+\frac{1}{2}} = E_z^n + C_b \delta_x^H H_y^n. \quad (53)$$

$$H_y^{n+\frac{1}{2}} = H_y^n + D_b \delta_x^E E_z^{n+\frac{1}{2}} + S_{H,1}^n. \quad (54)$$

$$E_z^{n+1} = E_z^{n+\frac{1}{2}} + C_b \delta_x^H H_y^{n+\frac{1}{2}}. \quad (55)$$

$$H_y^{n+1} = H_y^{n+\frac{1}{2}} + D_b \delta_x^E E_z^{n+\frac{1}{2}} + S_{H,2}^n. \quad (56)$$

Shifting back equations (55) and (56) one full step gives

$$E_z^n = E_z^{n-\frac{1}{2}} + C_b \delta_x^H H_y^{n-\frac{1}{2}}. \quad (57)$$

$$H_y^n = H_y^{n-\frac{1}{2}} + D_b \delta_x^E E_z^{n-\frac{1}{2}} + S_{H,2}^{n-1}. \quad (58)$$

To eliminate E_z from equations (54)–(56), equation (54) gives

$$D_b \delta_x^E E_z^{n+\frac{1}{2}} = H_y^{n+\frac{1}{2}} - H_y^n - S_{H,1}^n. \quad (59)$$

Substituting into equation (56) gives

$$H_y^{n+1} = 2H_y^{n+\frac{1}{2}} - H_y^n - S_{H,1}^n + S_{H,2}^n. \quad (60)$$

Solving for the half-step value gives

$$H_y^{n+\frac{1}{2}} = \frac{1}{2} (H_y^{n+1} + H_y^n + S_{H,1}^n - S_{H,2}^n). \quad (61)$$

Shifting back one full step gives

$$H_y^{n-\frac{1}{2}} = \frac{1}{2} (H_y^n + H_y^{n-1} + S_{H,1}^{n-1} - S_{H,2}^{n-1}). \quad (62)$$

Next, eliminate E_z using equations (53), (54), (57), and (58). Applying $D_b \delta_x^E$ to both sides of equation (57) gives

$$D_b \delta_x^E E_z^n = D_b \delta_x^E E_z^{n-\frac{1}{2}} + \gamma \delta_x^{H,2} H_y^{n-\frac{1}{2}}. \quad (63)$$

Eliminate $D_b \delta_x^E E_z^{n-\frac{1}{2}}$ by equation (58) gives

$$D_b \delta_x^E E_z^n = H_y^n - (1 - \gamma \delta_x^{H,2}) H_y^{n-\frac{1}{2}} - S_{H,2}^{n-1}. \quad (64)$$

Applying $D_b \delta_x^E$ to both sides of equation (53) gives

$$(1 - \gamma \delta_x^{H,2}) D_b \delta_x^E E_z^{n+\frac{1}{2}} = D_b \delta_x^E E_z^n + \gamma \delta_x^{H,2} H_y^n. \quad (65)$$

Eliminate $D_b \delta_x^E E_z^{n+\frac{1}{2}}$ by equation (54) gives

$$D_b \delta_x^E E_z^n = (1 - \gamma \delta_x^{H,2}) H_y^{n+\frac{1}{2}} - H_y^n - (1 - \gamma \delta_x^{H,2}) S_{H,1}^n. \quad (66)$$

Equating equations (66) and (64) eliminates E_z^n :

$$(1 - \gamma \delta_x^{H,2}) (H_y^{n+\frac{1}{2}} + H_y^{n-\frac{1}{2}}) = 2H_y^n + (1 - \gamma \delta_x^{H,2}) S_{H,1}^n - S_{H,2}^{n-1}. \quad (67)$$

Finally, substitute equations (61) and (62) into equation (67) to eliminate the half-step values of H_y :

$$(1 - \gamma \delta_x^{H,2}) H_y^{n+1} = 2(1 + \gamma \delta_x^{H,2}) H_y^n - (1 - \gamma \delta_x^{H,2}) H_y^{n-1} + (1 - \gamma \delta_x^{H,2}) S_{H,1}^n - (1 - \gamma \delta_x^{H,2}) S_{H,1}^{n-1} + (1 - \gamma \delta_x^{H,2}) S_{H,2}^n - (1 + \gamma \delta_x^{H,2}) S_{H,2}^{n-1}. \quad (68)$$

Recall that $\gamma = C_b D_b = c^2 \Delta t^2 / 4$, then equation (68) results in,

$$\frac{H_y^{n+1} - 2H_y^n + H_y^{n-1}}{\Delta t^2} = c^2 \delta_x^{H,2} \left(\frac{H_y^{n+1} + 2H_y^n + H_y^{n-1}}{4} \right) + \frac{1}{\Delta t^2} \left[(1 - \gamma \delta_x^{H,2}) S_{H,1}^n - (1 - \gamma \delta_x^{H,2}) S_{H,1}^{n-1} + (1 - \gamma \delta_x^{H,2}) S_{H,2}^n - (1 + \gamma \delta_x^{H,2}) S_{H,2}^{n-1} \right]. \quad (69)$$

Suppose the same source is applied in both substeps, with $S_{H,1}^n = S_{H,2}^n = \frac{\Delta t}{2} S_H^{n+\frac{1}{2}}$ and $S_{H,1}^{n-1} = S_{H,2}^{n-1} = \frac{\Delta t}{2} S_H^{n-\frac{1}{2}}$. Then equation (69) becomes

$$\frac{H_y^{n+1} - 2H_y^n + H_y^{n-1}}{\Delta t^2} = c^2 \delta_x^{H,2} \left(\frac{H_y^{n+1} + 2H_y^n + H_y^{n-1}}{4} \right) + \frac{(1 - \gamma \delta_x^{H,2}) S_H^{n+\frac{1}{2}} - S_H^{n-\frac{1}{2}}}{\Delta t}. \quad (70)$$

Assume $S_H^{n+a} = S_H(t_n + a\Delta t)$ and $F_H^n = (S_H)_t(t_n)$. The source approximation in equation (70) has the Taylor expansion

$$F_H^n - \frac{\gamma}{\Delta t} \delta_x^{H,2} S_H^n - \frac{\gamma}{2} \delta_x^{H,2} F_H^n + \frac{\Delta t^2}{24} (S_H)_{ttt}^n + O(\Delta t^3), \quad \text{error} = -\frac{\gamma}{\Delta t} \delta_x^{H,2} S_H^n - \frac{\gamma}{2} \delta_x^{H,2} F_H^n + \frac{\Delta t^2}{24} (S_H)_{ttt}^n + O(\Delta t^3). \quad (71)$$

Thus the two-substep midpoint-source case is first-order accurate for a spatially varying S_H , and second-order when $\delta_x^{H,2} S_H = 0$.

Alternatively, suppose the source is added only once in the first substep and evaluated at the midpoint, with $S_{H,1}^n = \Delta t S_H^{n+\frac{1}{2}}$, $S_{H,2}^n = 0$, $S_{H,1}^{n-1} = \Delta t S_H^{n-\frac{1}{2}}$, and $S_{H,2}^{n-1} = 0$. Then equation (69) becomes

$$\frac{H_y^{n+1} - 2H_y^n + H_y^{n-1}}{\Delta t^2} = c^2 \delta_x^{H,2} \left(\frac{H_y^{n+1} + 2H_y^n + H_y^{n-1}}{4} \right) + (1 - \gamma \delta_x^{H,2}) \frac{S_H^{n+\frac{1}{2}} - S_H^{n-\frac{1}{2}}}{\Delta t}. \quad (72)$$

The source approximation in equation (72) has the Taylor expansion

$$F_H^n - \gamma \delta_x^{H,2} F_H^n + \frac{\Delta t^2}{24} (S_H)_{ttt}^n + O(\Delta t^4), \quad \text{error} = -\gamma \delta_x^{H,2} F_H^n + \frac{\Delta t^2}{24} (S_H)_{ttt}^n + O(\Delta t^4). \quad (73)$$

Since $\gamma = O(\Delta t^2)$, the single midpoint-source case is second-order accurate for $F_H^n = (S_H)_t(t_n)$.

Alternatively, suppose the first source is evaluated at the first quarter-step and the second source at the third quarter-step: $S_{H,1}^n = \frac{\Delta t}{2} S_H^{n+\frac{1}{4}}$, $S_{H,2}^n = \frac{\Delta t}{2} S_H^{n+\frac{3}{4}}$, $S_{H,1}^{n-1} = \frac{\Delta t}{2} S_H^{n-\frac{3}{4}}$, and $S_{H,2}^{n-1} = \frac{\Delta t}{2} S_H^{n-\frac{1}{4}}$. Then equation (69) becomes

$$\begin{aligned} & \frac{H_y^{n+1} - 2H_y^n + H_y^{n-1}}{\Delta t^2} = \\ & c^2 \delta_x^{H,2} \left(\frac{H_y^{n+1} + 2H_y^n + H_y^{n-1}}{4} \right) + \frac{1}{2\Delta t} \left[(1 - \gamma \delta_x^{H,2}) S_H^{n+\frac{1}{4}} - (1 - \gamma \delta_x^{H,2}) S_H^{n-\frac{3}{4}} + (1 - \gamma \delta_x^{H,2}) S_H^{n+\frac{3}{4}} - (1 + \gamma \delta_x^{H,2}) S_H^{n-\frac{1}{4}} \right]. \end{aligned} \quad (74)$$

The source approximation in equation (74) has the Taylor expansion

$$F_H^n - \frac{\gamma}{\Delta t} \delta_x^{H,2} S_H^n - \frac{3\gamma}{4} \delta_x^{H,2} F_H^n + \frac{7\Delta t^2}{96} (S_H)_{ttt}^n + O(\Delta t^3), \quad \text{error} = -\frac{\gamma}{\Delta t} \delta_x^{H,2} S_H^n - \frac{3\gamma}{4} \delta_x^{H,2} F_H^n + \frac{7\Delta t^2}{96} (S_H)_{ttt}^n + O(\Delta t^3). \quad (75)$$

Thus the quarter-step source case is first-order accurate for a spatially varying S_H , and second-order when $\delta_x^{H,2} S_H = 0$.

Assume $S_H(x, t) = \hat{S}_H e^{i(\omega t - kx)}$, and define $\theta = \omega \Delta t$, $\delta_x^{H,2} e^{-ikx} = \lambda_{H,x} e^{-ikx}$, and $q_H = -\gamma \lambda_{H,x} \geq 0$. Substituting into the source term on the right-hand side of equation (70) gives

$$\begin{aligned} F_{H,\text{mid},2}^n &= \frac{(1 - \gamma \delta_x^{H,2}) S_H^{n+\frac{1}{2}} - S_H^{n-\frac{1}{2}}}{\Delta t} \\ &= \frac{\hat{S}_H e^{i(\omega t_n - kx)}}{\Delta t} \left[(1 + q_H) e^{i\theta/2} - e^{-i\theta/2} \right] \\ &= \frac{\hat{S}_H}{\Delta t} \left[q_H \cos\left(\frac{\theta}{2}\right) + i(2 + q_H) \sin\left(\frac{\theta}{2}\right) \right] e^{i(\omega t_n - kx)}. \end{aligned} \quad (76)$$

so $\hat{F}_{H,\text{mid},2} = \frac{\hat{S}_H}{\Delta t} [q_H \cos(\frac{\theta}{2}) + i(2 + q_H) \sin(\frac{\theta}{2})]$. Substituting into the source term on the right-hand side of equation (72) gives

$$\begin{aligned} F_{H,\text{mid},1}^n &= (1 - \gamma \delta_x^{H,2}) \frac{S_H^{n+\frac{1}{2}} - S_H^{n-\frac{1}{2}}}{\Delta t} \\ &= \frac{(1 + q_H) \hat{S}_H e^{i(\omega t_n - kx)}}{\Delta t} (e^{i\theta/2} - e^{-i\theta/2}) \\ &= \frac{2i(1 + q_H) \hat{S}_H}{\Delta t} \sin\left(\frac{\theta}{2}\right) e^{i(\omega t_n - kx)}. \end{aligned} \quad (77)$$

so $\hat{F}_{H,\text{mid},1} = \frac{2i(1+q_H)\hat{S}_H}{\Delta t} \sin(\frac{\theta}{2})$. Substituting into the source term on the right-hand side of equation (74) gives

$$\begin{aligned} F_{H,\text{quarter}}^n &= \frac{1}{2\Delta t} \left[(1 - \gamma \delta_x^{H,2}) S_H^{n+\frac{1}{4}} - (1 - \gamma \delta_x^{H,2}) S_H^{n-\frac{3}{4}} + (1 - \gamma \delta_x^{H,2}) S_H^{n+\frac{3}{4}} - (1 + \gamma \delta_x^{H,2}) S_H^{n-\frac{1}{4}} \right] \\ &= \frac{\hat{S}_H e^{i(\omega t_n - kx)}}{2\Delta t} \left[(1 + q_H) e^{i\theta/4} - (1 + q_H) e^{-3i\theta/4} + (1 + q_H) e^{3i\theta/4} - (1 - q_H) e^{-i\theta/4} \right] \\ &= \frac{\hat{S}_H}{\Delta t} \left[q_H \cos\left(\frac{\theta}{4}\right) + i \left(\sin\left(\frac{\theta}{4}\right) + (1 + q_H) \sin\left(\frac{3\theta}{4}\right) \right) \right] e^{i(\omega t_n - kx)}. \end{aligned} \quad (78)$$

Therefore the magnitude ratios between the quarter-step and midpoint magnetic-source terms are

$$\frac{|\hat{F}_{H,\text{quarter}}|}{|\hat{F}_{H,\text{mid},2}|} = \frac{\sqrt{q_H^2 \cos^2(\frac{\theta}{4}) + [\sin(\frac{\theta}{4}) + (1 + q_H) \sin(\frac{3\theta}{4})]^2}}{\sqrt{q_H^2 \cos^2(\frac{\theta}{2}) + (2 + q_H)^2 \sin^2(\frac{\theta}{2})}}, \quad (79)$$

$$\frac{|\hat{F}_{H,\text{quarter}}|}{|\hat{F}_{H,\text{mid},1}|} = \frac{\sqrt{q_H^2 \cos^2(\frac{\theta}{4}) + [\sin(\frac{\theta}{4}) + (1 + q_H) \sin(\frac{3\theta}{4})]^2}}{2(1 + q_H) |\sin(\frac{\theta}{2})|}. \quad (80)$$

For a spatially uniform source, $q_H = 0$, both midpoint placements give the same magnitude, and the quarter-step magnitude is smaller by the factor

$$\frac{|\hat{F}_{H,\text{quarter}}|}{|\hat{F}_{H,\text{mid},2}|} = \frac{|\hat{F}_{H,\text{quarter}}|}{|\hat{F}_{H,\text{mid},1}|} = \left| \cos\left(\frac{\omega \Delta t}{4}\right) \right|. \quad (81)$$

2.4 Method of manufactured solution validation for magnetic source

To test the magnetic-source placements of the previous section with a genuinely nonzero forcing, consider the one-dimensional wave equation

$$H_{tt} - c^2 H_{xx} = f(x, t), \quad x \in [0, L], \quad (82)$$

with PEC boundary condition, $E(0, t) = E(L, t) = 0$. Choose a smooth manufactured solution,

$$H_{\text{MMS}}(x, t) = \cos(kx) \cos(\omega t), \quad k = \frac{2\pi}{L}, \quad \omega = \frac{3\pi c}{L}, \quad (83)$$

where we set $\omega \neq ck$ in order to have a non-zero f .

Indeed,

$$H_{tt} = -\omega^2 \cos(kx) \cos(\omega t), \quad H_{xx} = -k^2 \cos(kx) \cos(\omega t), \quad (84)$$

so

$$f(x, t) = (c^2 k^2 - \omega^2) \cos(kx) \cos(\omega t) = -\frac{5\pi^2 c^2}{L^2} \cos(kx) \cos(\omega t). \quad (85)$$

In the ADI update, $\partial_t H = \mu^{-1} \partial_x E + S_H$ with $F_H = (S_H)_t$ the wave-equation source of (69). Matching $F_H = f$ therefore gives the continuous magnetic soft source

$$S_H(x, t) = \frac{c^2 k^2 - \omega^2}{\omega} \cos(kx) \sin(\omega t). \quad (86)$$

From (83), we have:

$$H(x, 0) = \cos(kx), \quad H_t(x, 0) = 0, \quad (87)$$

and the compatible electric field is $E(x, 0) = 0$ (since $S_H(x, 0) = 0$ as well). The wavenumber $k = 2\pi/L$ is compatible with the PEC conditions through $E \propto \sin(kx)$.

To analyse how the magnitude will scale with respect to the CFL numbers, we model the system as a forced oscillation system with natural frequency $\omega_{\text{ADI}}(k)$ and a forcing function whose frequency is $\omega_d = \omega$ from (83).

We model the system as a simple undamped forced oscillator,

$$a'' + \omega_{\text{ADI}}^2 a = F_H(t). \quad (88)$$

First, we consider $F_H(t) = F_{H, \text{mid}, 2}$ from (76). On the mode $\cos(kx)$ one has $\delta_x^{H, 2} \cos(kx) = \lambda_{H, x} \cos(kx)$ and $q_H = -\gamma \lambda_{H, x}$, so with (86),

$$\begin{aligned} F_{H, \text{mid}, 2} &= \frac{1}{\Delta t} \left[(1 - \gamma \delta_x^{H, 2}) S_H^{n+\frac{1}{2}} - S_H^{n-\frac{1}{2}} \right] \\ &= \frac{c^2 k^2 - \omega^2}{\omega \Delta t} \cos(kx) \left[(1 + q_H) \sin(\omega(t_n + \tfrac{1}{2} \Delta t)) - \sin(\omega(t_n - \tfrac{1}{2} \Delta t)) \right] \\ &= \frac{c^2 k^2 - \omega^2}{\omega \Delta t} \cos(kx) \left[(2 + q_H) \sin\left(\frac{\omega \Delta t}{2}\right) \cos(\omega t_n) + q_H \cos\left(\frac{\omega \Delta t}{2}\right) \sin(\omega t_n) \right]. \end{aligned} \quad (89)$$

The corresponding modal system is

$$\begin{aligned} a'' + \omega_{\text{ADI}}^2 a &= \frac{c^2 k^2 - \omega^2}{\omega \Delta t} \left[(2 + q_H) \sin\left(\frac{\omega \Delta t}{2}\right) \cos(\omega t) + q_H \cos\left(\frac{\omega \Delta t}{2}\right) \sin(\omega t) \right], \\ a(0) &= 1, \\ a'(0) &= 0. \end{aligned} \quad (90)$$

Writing the drive as $A \cos(\omega t) + B \sin(\omega t)$ with

$$A = \frac{c^2 k^2 - \omega^2}{\omega \Delta t} (2 + q_H) \sin\left(\frac{\omega \Delta t}{2}\right), \quad B = \frac{c^2 k^2 - \omega^2}{\omega \Delta t} q_H \cos\left(\frac{\omega \Delta t}{2}\right), \quad (91)$$

the solution is

$$\begin{aligned} a(t) &= \left(1 - \frac{A}{\omega_{\text{ADI}}^2 - \omega^2} \right) \cos(\omega_{\text{ADI}} t) - \frac{\omega}{\omega_{\text{ADI}}} \frac{B}{\omega_{\text{ADI}}^2 - \omega^2} \sin(\omega_{\text{ADI}} t) \\ &\quad + \frac{A}{\omega_{\text{ADI}}^2 - \omega^2} \cos(\omega t) + \frac{B}{\omega_{\text{ADI}}^2 - \omega^2} \sin(\omega t). \end{aligned} \quad (92)$$

Next, consider $F_H(t) = F_{H,\text{mid},1}$ from (77). With the manufactured soft source (86),

$$\begin{aligned}
F_{H,\text{mid},1} &= (1 - \gamma\delta_x^{H,2}) \frac{S_H^{n+\frac{1}{2}} - S_H^{n-\frac{1}{2}}}{\Delta t} \\
&= \frac{c^2 k^2 - \omega^2}{\omega} (1 + q_H) \cos(kx) \frac{\sin(\omega(t_n + \frac{1}{2}\Delta t)) - \sin(\omega(t_n - \frac{1}{2}\Delta t))}{\Delta t} \\
&= \frac{c^2 k^2 - \omega^2}{\omega} (1 + q_H) \cos(kx) \frac{2}{\Delta t} \sin\left(\frac{\omega\Delta t}{2}\right) \cos(\omega t_n).
\end{aligned} \tag{93}$$

Thus corresponding system is

$$\begin{aligned}
a'' + \omega_{\text{ADI}}^2 a &= \frac{2}{\Delta t} \sin\left(\frac{\omega\Delta t}{2}\right) \cdot (1 + q_H) \frac{c^2 k^2 - \omega^2}{\omega} \cos(\omega t), \\
a(0) &= 1, \\
a'(0) &= 0.
\end{aligned} \tag{94}$$

Solving this system yields

$$\begin{aligned}
a(t) &= (1 - A_p) \cos(\omega_{\text{ADI}} t) + A_p \cos(\omega t), \\
A_p &= \text{sinc}\left(\frac{\omega\Delta t}{2}\right) (1 + q_H) \frac{c^2 k^2 - \omega^2}{\omega_{\text{ADI}}^2 - \omega^2}.
\end{aligned} \tag{95}$$

Finally, consider $F_H(t) = F_{H,\text{quarter}}$ from (78). With (86),

$$\begin{aligned}
F_{H,\text{quarter}} &= \frac{1}{2\Delta t} \left[(1 - \gamma\delta_x^{H,2}) S_H^{n+\frac{1}{4}} - (1 - \gamma\delta_x^{H,2}) S_H^{n-\frac{3}{4}} \right. \\
&\quad \left. + (1 - \gamma\delta_x^{H,2}) S_H^{n+\frac{3}{4}} - (1 + \gamma\delta_x^{H,2}) S_H^{n-\frac{1}{4}} \right] \\
&= \frac{c^2 k^2 - \omega^2}{\omega\Delta t} \cos(kx) \left[\left(\sin\left(\frac{\omega\Delta t}{4}\right) + (1 + q_H) \sin\left(\frac{3\omega\Delta t}{4}\right) \right) \cos(\omega t_n) + q_H \cos\left(\frac{\omega\Delta t}{4}\right) \sin(\omega t_n) \right].
\end{aligned} \tag{96}$$

The corresponding modal system is

$$\begin{aligned}
a'' + \omega_{\text{ADI}}^2 a &= \frac{c^2 k^2 - \omega^2}{\omega\Delta t} \left[\left(\sin\left(\frac{\omega\Delta t}{4}\right) + (1 + q_H) \sin\left(\frac{3\omega\Delta t}{4}\right) \right) \cos(\omega t) + q_H \cos\left(\frac{\omega\Delta t}{4}\right) \sin(\omega t) \right], \\
a(0) &= 1, \\
a'(0) &= 0.
\end{aligned} \tag{97}$$

Writing the drive as $A \cos(\omega t) + B \sin(\omega t)$ with

$$A = \frac{c^2 k^2 - \omega^2}{\omega\Delta t} \left[\sin\left(\frac{\omega\Delta t}{4}\right) + (1 + q_H) \sin\left(\frac{3\omega\Delta t}{4}\right) \right], \quad B = \frac{c^2 k^2 - \omega^2}{\omega\Delta t} q_H \cos\left(\frac{\omega\Delta t}{4}\right), \tag{98}$$

the solution is

$$\begin{aligned}
a(t) &= \left(1 - \frac{A}{\omega_{\text{ADI}}^2 - \omega^2} \right) \cos(\omega_{\text{ADI}} t) - \frac{\omega}{\omega_{\text{ADI}}} \frac{B}{\omega_{\text{ADI}}^2 - \omega^2} \sin(\omega_{\text{ADI}} t) \\
&\quad + \frac{A}{\omega_{\text{ADI}}^2 - \omega^2} \cos(\omega t) + \frac{B}{\omega_{\text{ADI}}^2 - \omega^2} \sin(\omega t).
\end{aligned} \tag{99}$$

Relative to mid,2, the particular amplitudes satisfy the same magnitude ratios as (79)–(80).